



Modeling System Requirements

The Traditional Approach (DFD)

The Object Oriented Approach(UML Models)

FUNCTIONAL MODEL

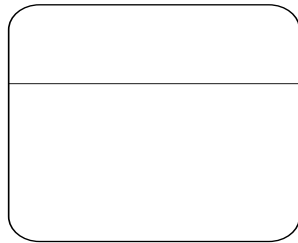
- ± Deals with the actual processes inside a system
- ± Is related to portion of *what happened* ?
- ± **Does not concern itself with how** or when something happens, but describes what has happened.
- ± Specifies meaning of processes in object model as well the actions in dynamic model.
- ± Most common way to describe functional model is to draw **Data Flow Diagram (DFD)**.

DATA FLOW DIAGRAMS [DFD'S]

± A diagram to show how data is captured, processed , stored and distributed within a system.

± This is generally represented during the analysis stage of a project, but can be further refined during the design stage to show more detail of how the system functions.

DFD Symbols (Gane & Sarson)



Process



Data Flow

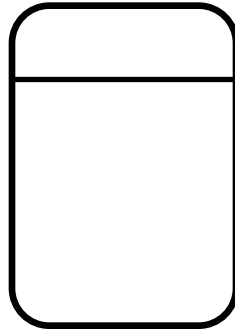


Data Store



Source/Sink (External Entity)

SYMBOLS USED IN A DFD



± **PROCESS**

Actions performed on data so that they are transformed, stored or distributed. This can be a computerised or manual transformation



± **DATA FLOW**

E.g. Result of a query to a database, contents of a printed report : data that moves together to common destinations

SYMBOLS USED IN A DFD

$\pm$ DATA STORE



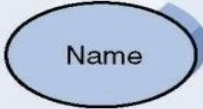
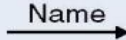
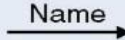
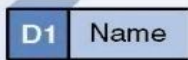
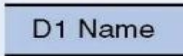
A physical location to hold data
e.g. a file folder or notebook etc.

$\pm$ SOURCE/SINK

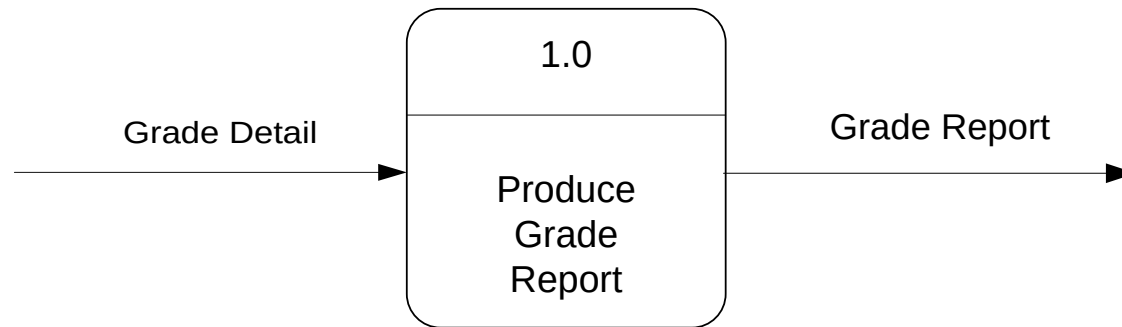


Origin and/or destination of the
data. Often referred to as external
entities as they are outside the
system. We are only interested in
how data flows in to or from the
system to them.

DFD Elements

Data Flow Diagram Element	Typical Computer-Aided Software Engineering Fields	Gane and Sarson Symbol	DeMarco and Yourdan Symbol
Every <i>process</i> has - A number - A name (verb phase) - A description - One or more output data flows - Usually one or more input data flows	Label (name) Type (process) Description (what is it) Process number Process description (Structured English) Notes		
Every <i>data flow</i> has - A name (a noun) - A description - One or more connections to a process	Label (name) Type (flow) Description Alias (another name) Composition (description of data elements) Notes		
Every <i>data store</i> has - A number - A name (a noun) - A description - One or more input data flows - Usually one or more output data flows	Label (name) Type (store) Description Alias (another name) Composition (description of data elements) Notes		
Every <i>external entity</i> has - A name (a noun) - A description	Label (name) Type (entity) Description Alias (another name) Entity description Notes		

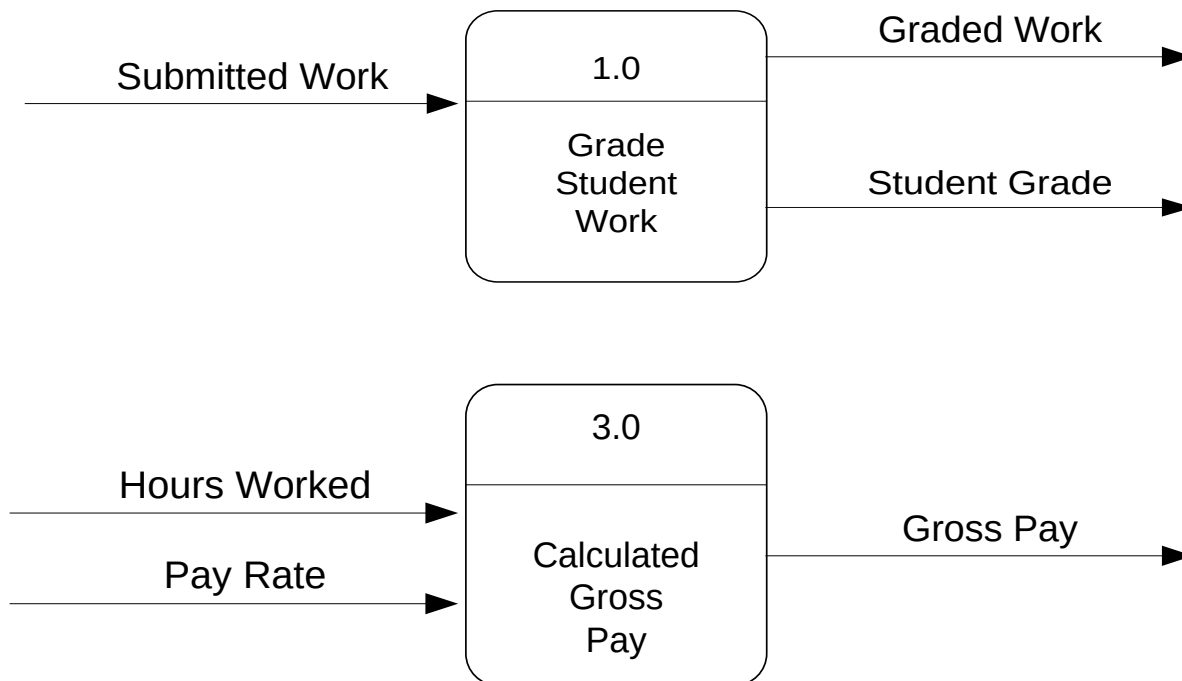
Process



- Work or actions performed on data (inside the system)
- Labels should be verb phrases
- Receives input data and produces output

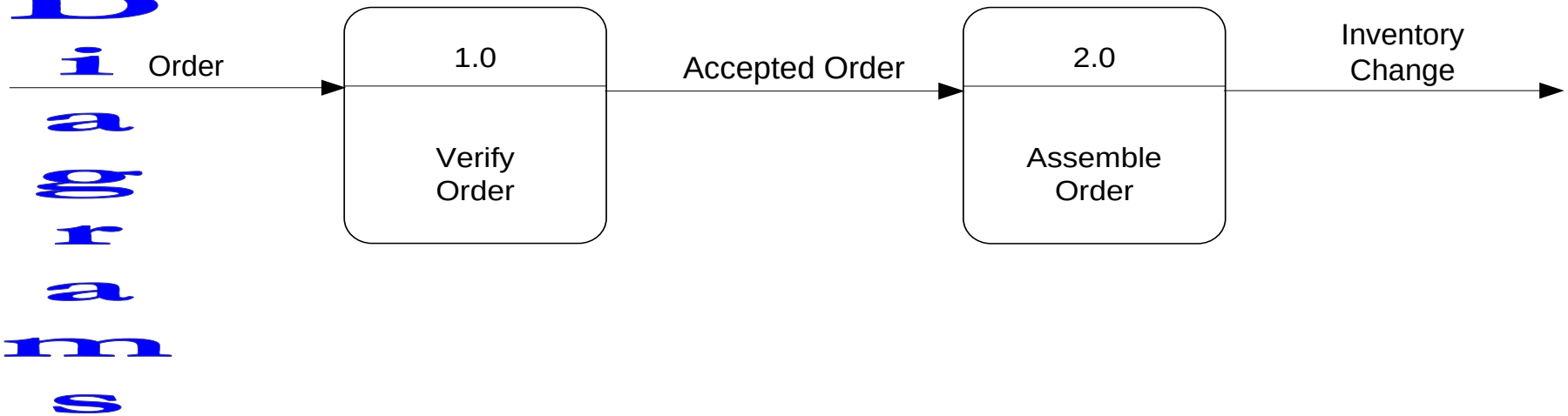
Rule 1: Process

- Can have more than one outgoing data flow or more than one incoming data flow



Rule 2: Process

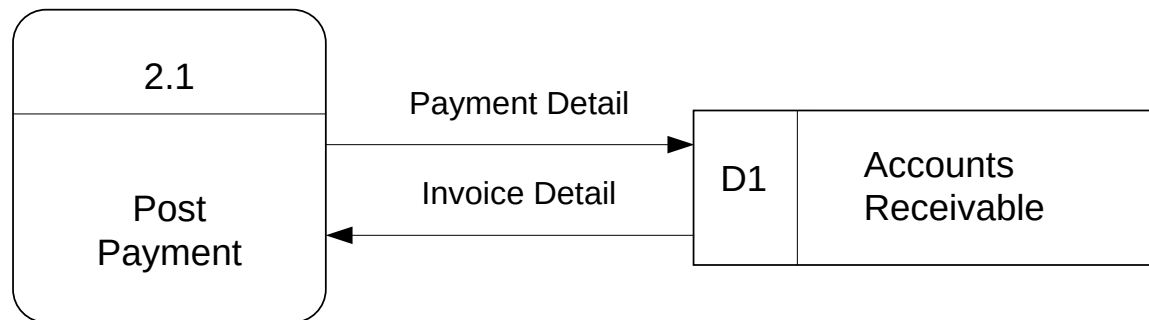
- Can connect to any other symbol (including another process symbol)



Data Flow



- Is a path for data to move from one part of the IS to another
- Arrows depicting movement of data
- Can represent flow between process and data store by two separate arrows



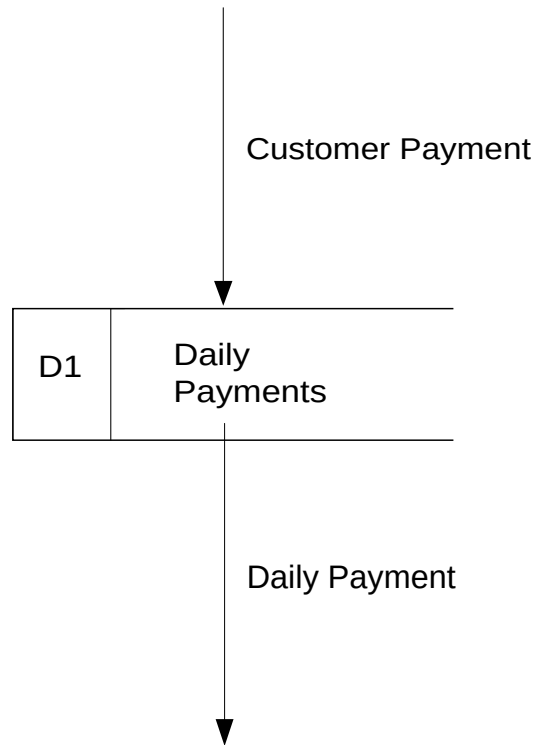
Data Store

D1	Students
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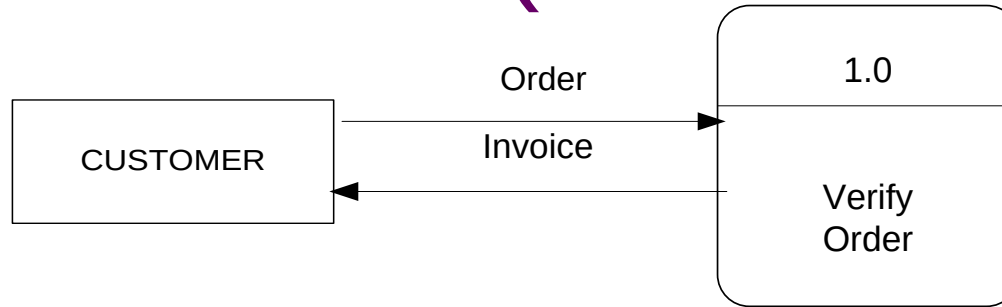
- Is used in a DFD to represent data that the system stores
- Labels should be noun phrases

Rule: Data Store

- Must have at least one incoming and one outgoing data flow



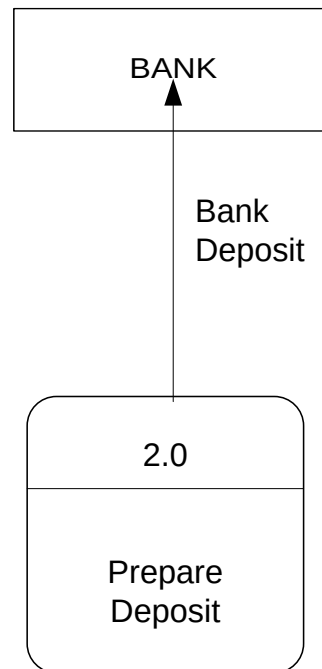
Source/Sink (External Entity)



- External entity that is origin or destination of data (outside the system)
- Is the singular form of a department, outside organisation, other IS, or person
- Labels should be noun phrases
- Source – Entity that supplies data to the system
- Sink – Entity that receives data from the system

Rule: Source/Sink

- Must be connected to a process by a data flow



Data Flow Diagrams

Rules for Using DFD Symbols

Data Flow That Connects

	YES	NO
A process to another process	✓	
A process to an external entity	✓	
A process to a data store	✓	
An external entity to another external entity		✓
An external entity to a data store		✓
A data store to another data store		✓

How Much Detail ?

± It is often impossible to represent a complete business on one diagram. For this reason it is quite common to use more than one level of Data Flow Diagram with each level showing more detailed information about part of the previous diagram.

± It is important that the diagram should be made as clear as possible and for this reason it is not uncommon to draw the same source or sink [external entity] more than once to help achieve this.

Context or Level 0 Diagram

The highest level view of a system. This only has one process which represents the overall function of the system and has no data stores as all the data is stored within the process.

Main steps when constructing a Level 0 Diagram :

- ± Identify where data is captured from
- ± Identify where data is distributed to
- ± Describe the overall process
- ± Map these out in a diagram using the correct symbols
- ± Link them with data flows that are labelled

Constructing a Level 1 diagram

- ± Identify and draw the processes that make up the Level 0 process
- ± Allocate descriptions to these
- ± Lay out the sources/sinks and data flows from the Level 0 diagram
- ± Draw in any data stores used in the process
- ± Link the new processes and data stores with named data links

Constructing a Level 2/3 diagram

IF it is necessary to produce a level 2 or level 3 diagram, the same steps would be followed as used when constructing the Level 1 diagram



Here is an example of how Data flow diagrams would be used to model the logic of data flows in a fast food burger bar.

1. Context or Level 0 Diagram

- ± Identify where data is captured from**
- ± Identify where data is distributed to
- ± Describe the overall process
- ± Map these out in a diagram using the correct symbols
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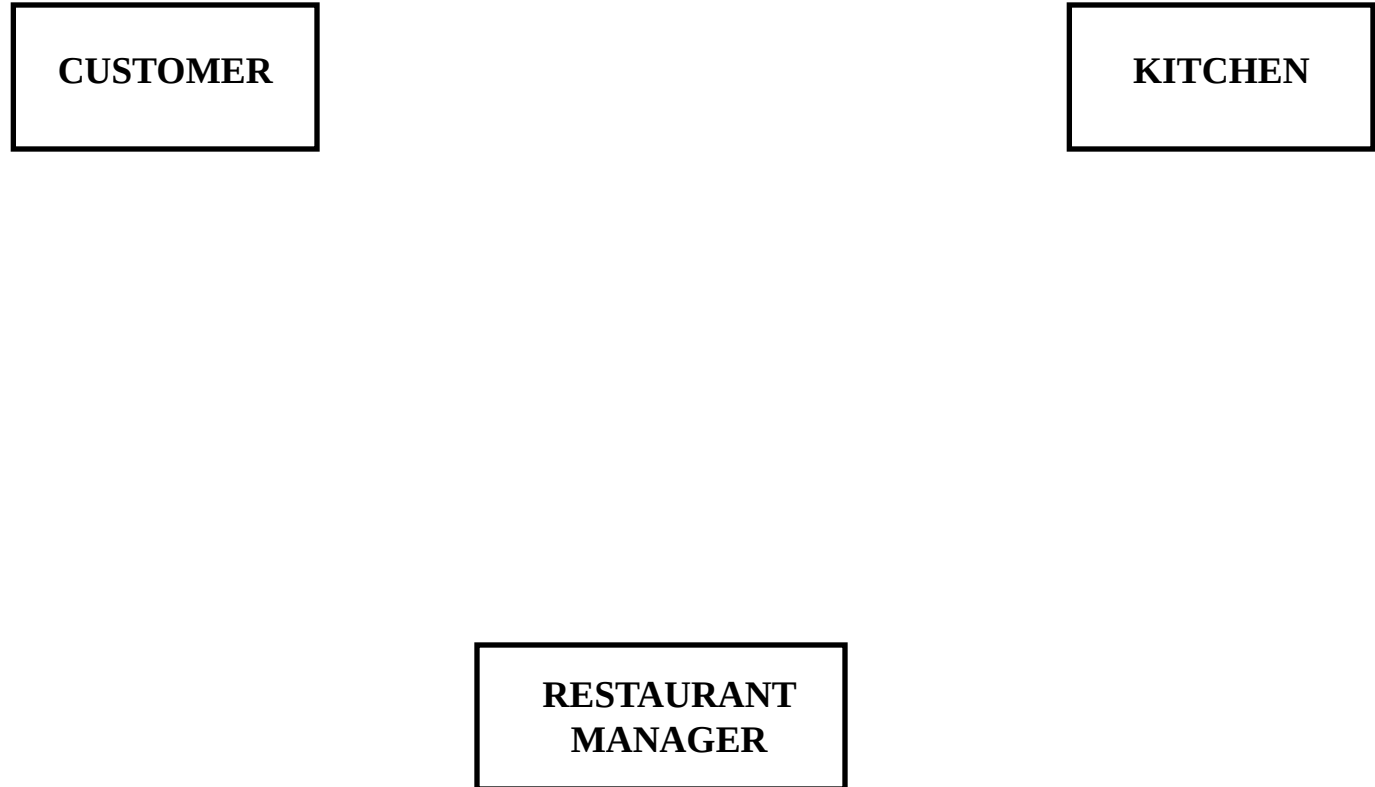
Data Flow Diagrams

Level 0 Context diagram



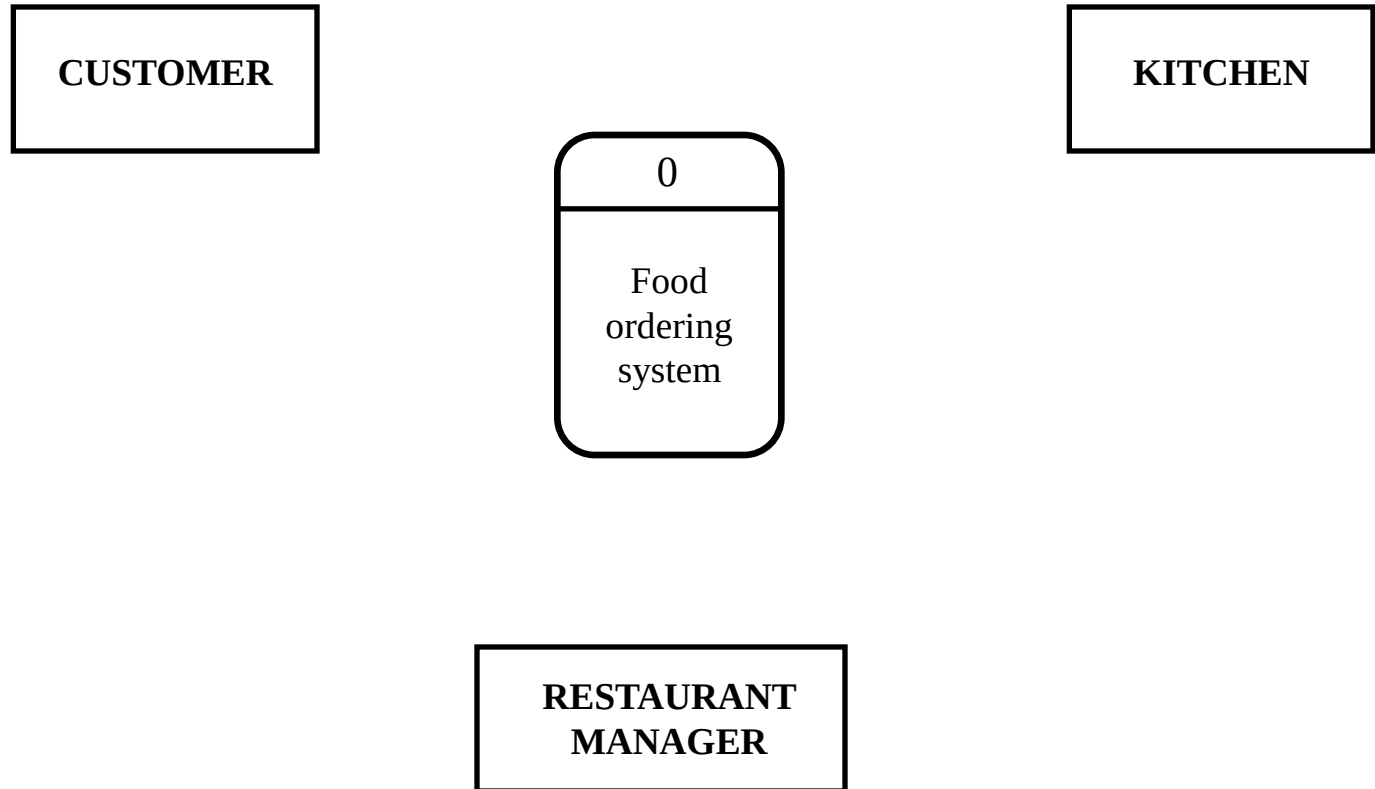
- ± Identify where data is captured from
- ± **Identify where data is distributed to**
- ± Describe the overall process
- ± Map these out in a diagram using the correct symbols
- ± Link them with data flows that are labelled

Level 0 Context diagram



- ± Identify where data is captured from
- ± Identify where data is distributed to
- ± **Describe the overall process**
- ± **Map these out in a diagram using the correct symbols**
- ± Link them with data flows that are labelled

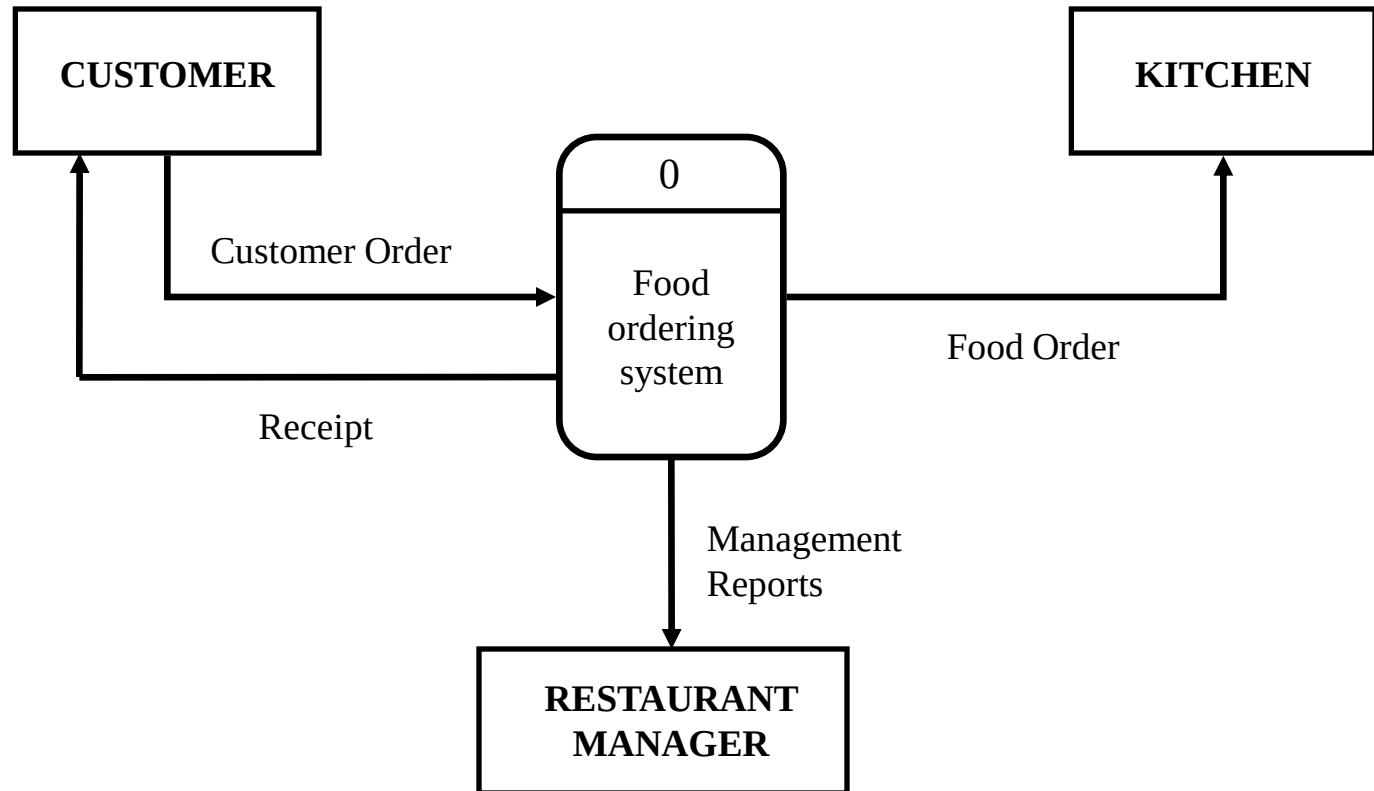
Level 0 Context diagram



How to draw a Data Flow Diagram

- ± Identify where data is captured from
- ± Identify where data is distributed to
- ± Describe the overall process
- ± Map these out using the correct symbols
- ± **Link them with data flows that are labelled**

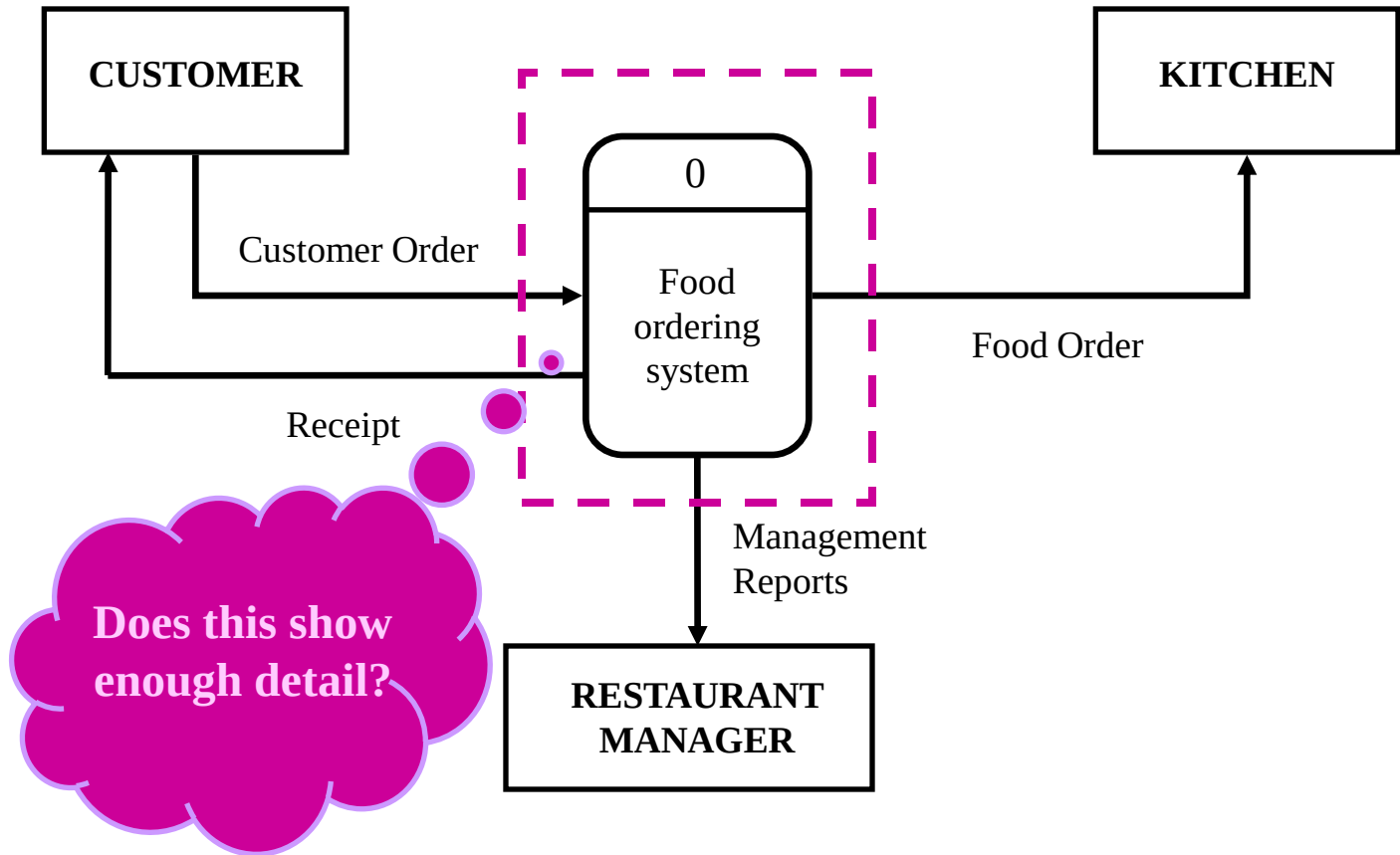
Level 0 Context diagram



Constructing a Level 1 diagram

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Level 0 Context diagram



Constructing a Level 1 diagram

± Identify and draw the processes that make up the Level 0 process

± Allocate descriptions to these

± Lay out the sources/sinks and data flows from the Level 0 diagram

± Draw in any data stores used in the process

± Link the new processes and data stores with named data links

Level 1 diagram

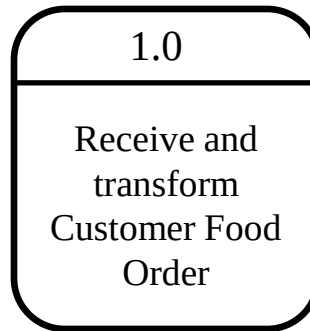
These new processes may show a process which corresponds to actions such as :

± Capturing data from different sources

± Maintaining data stores

± Producing and distributing data to different sinks

Share Data Workflow Diagram



Level 1 diagram

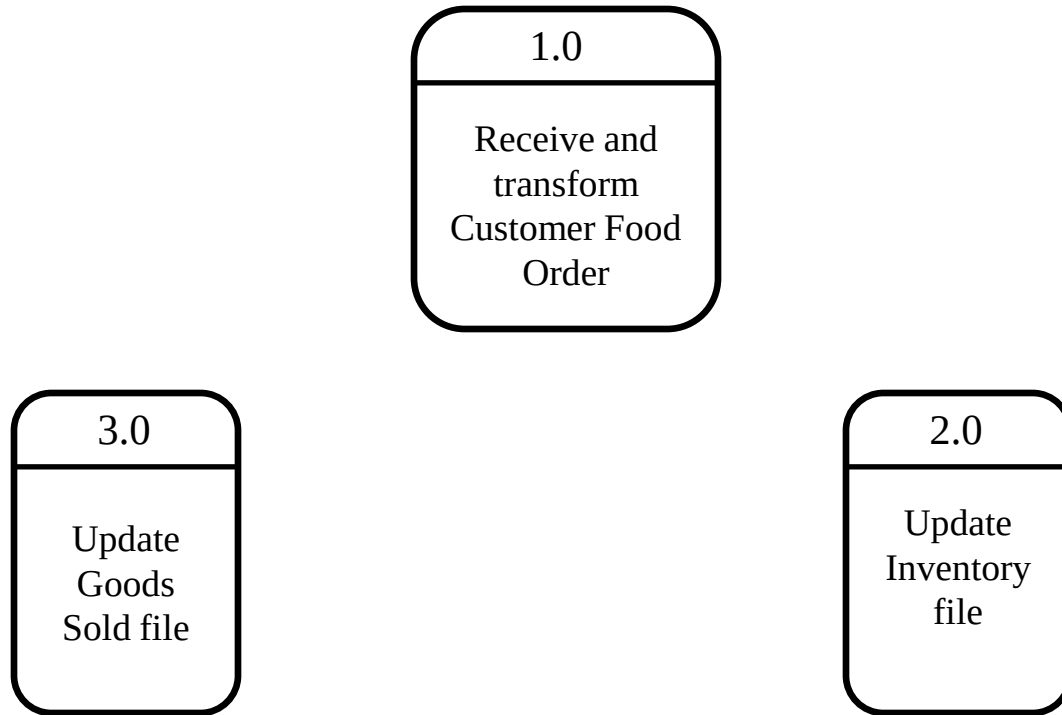
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± **Maintaining data stores**

± Producing and distributing data to different sinks

Database for Food Service

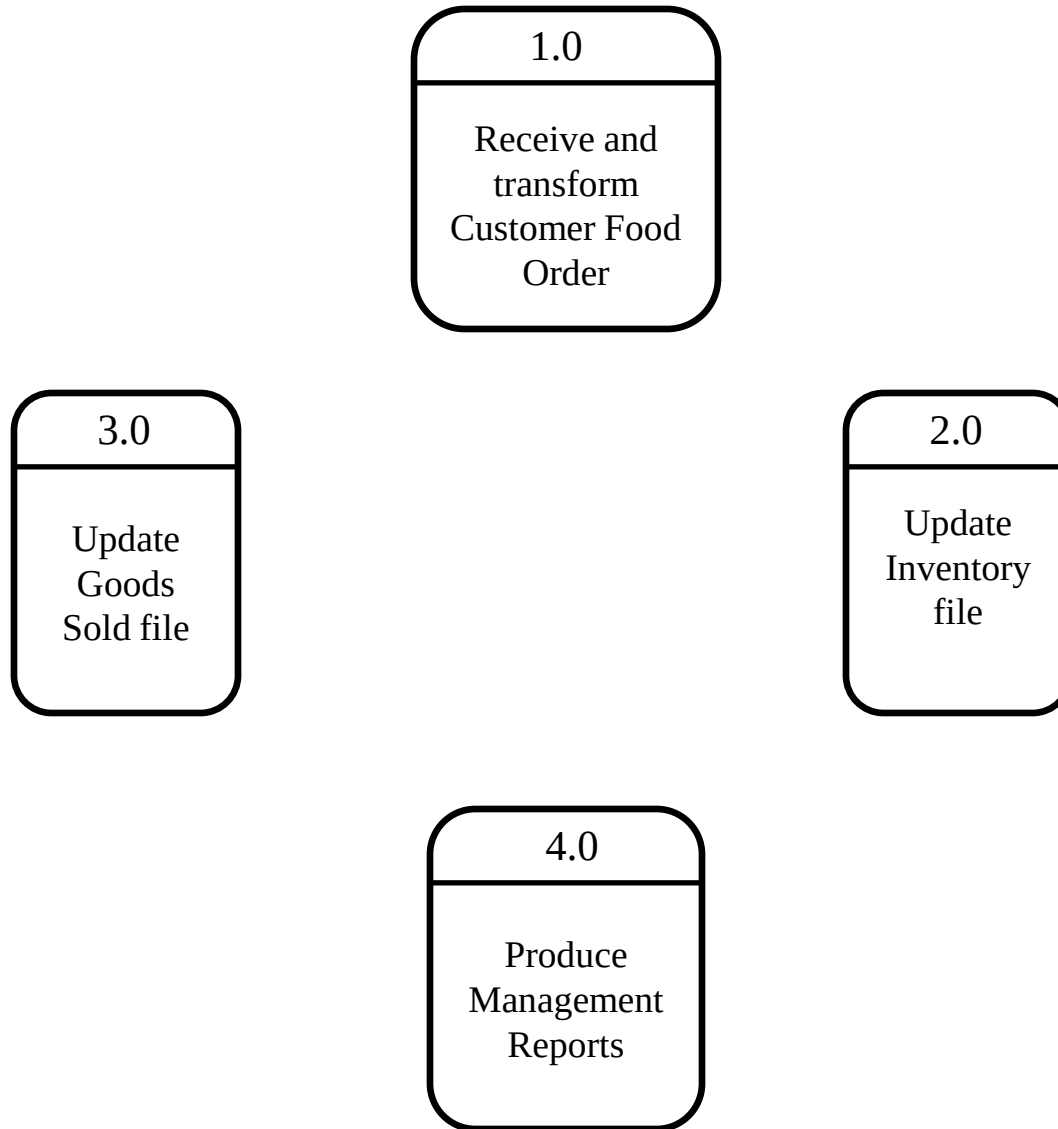


Level 1 diagram

These may show a process which corresponds to actions such as :

- ± Capturing data from different sources
- ± Maintaining data stores
- ± Producing and distributing data to different sinks

Database for Food Service



Constructing a Level 1 diagram

± Identify and draw the processes that make up the Level 0 process

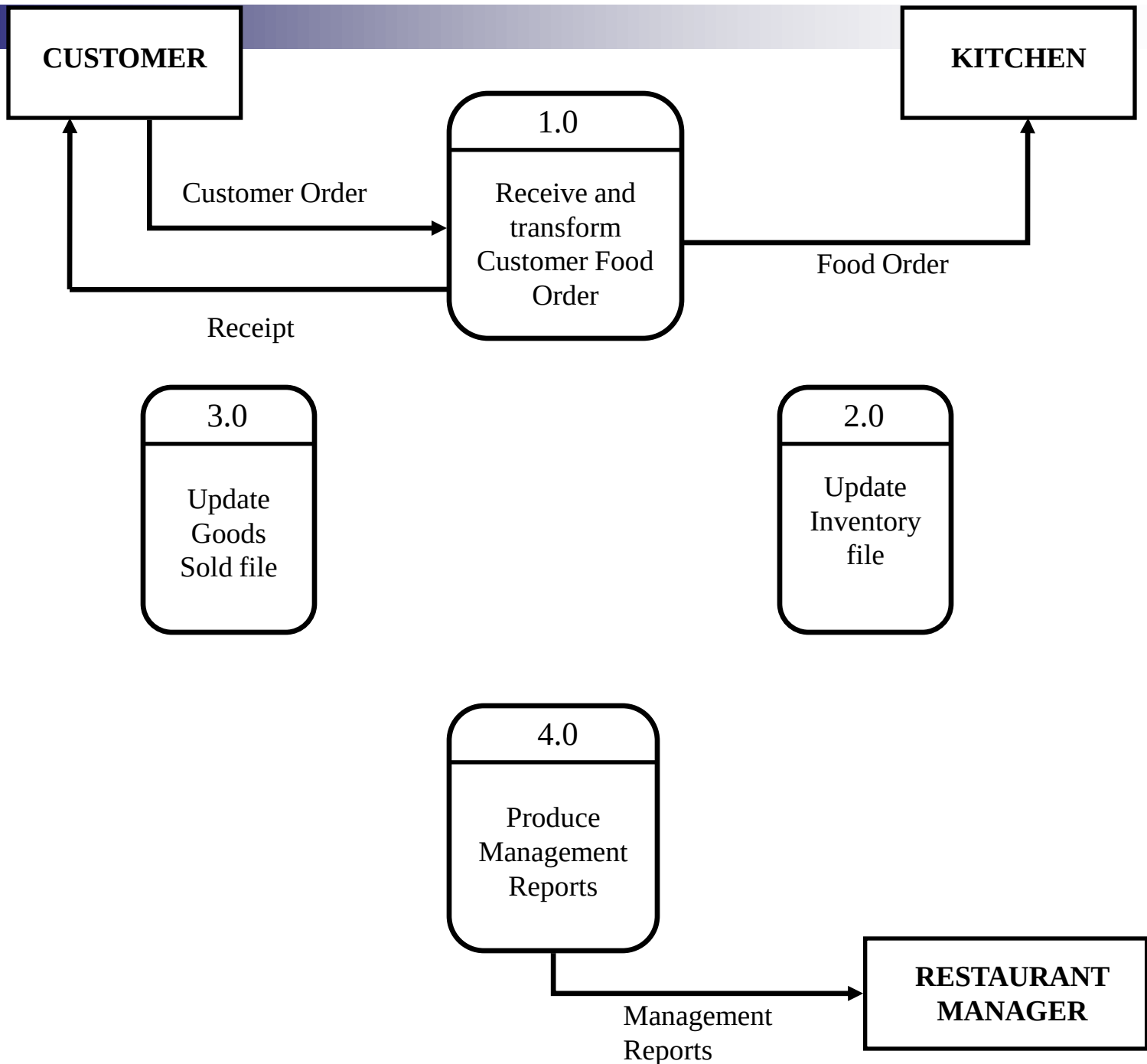
± Allocate descriptions to these

± **Lay out the sources/sinks and data flows from the Level 0 diagram**

± Draw in any data stores used in the process

± Link the new processes and data stores with named data links

Data Flow Diagrams



Constructing a Level 1 diagram

± Identify and draw the processes that make up the Level 0 process

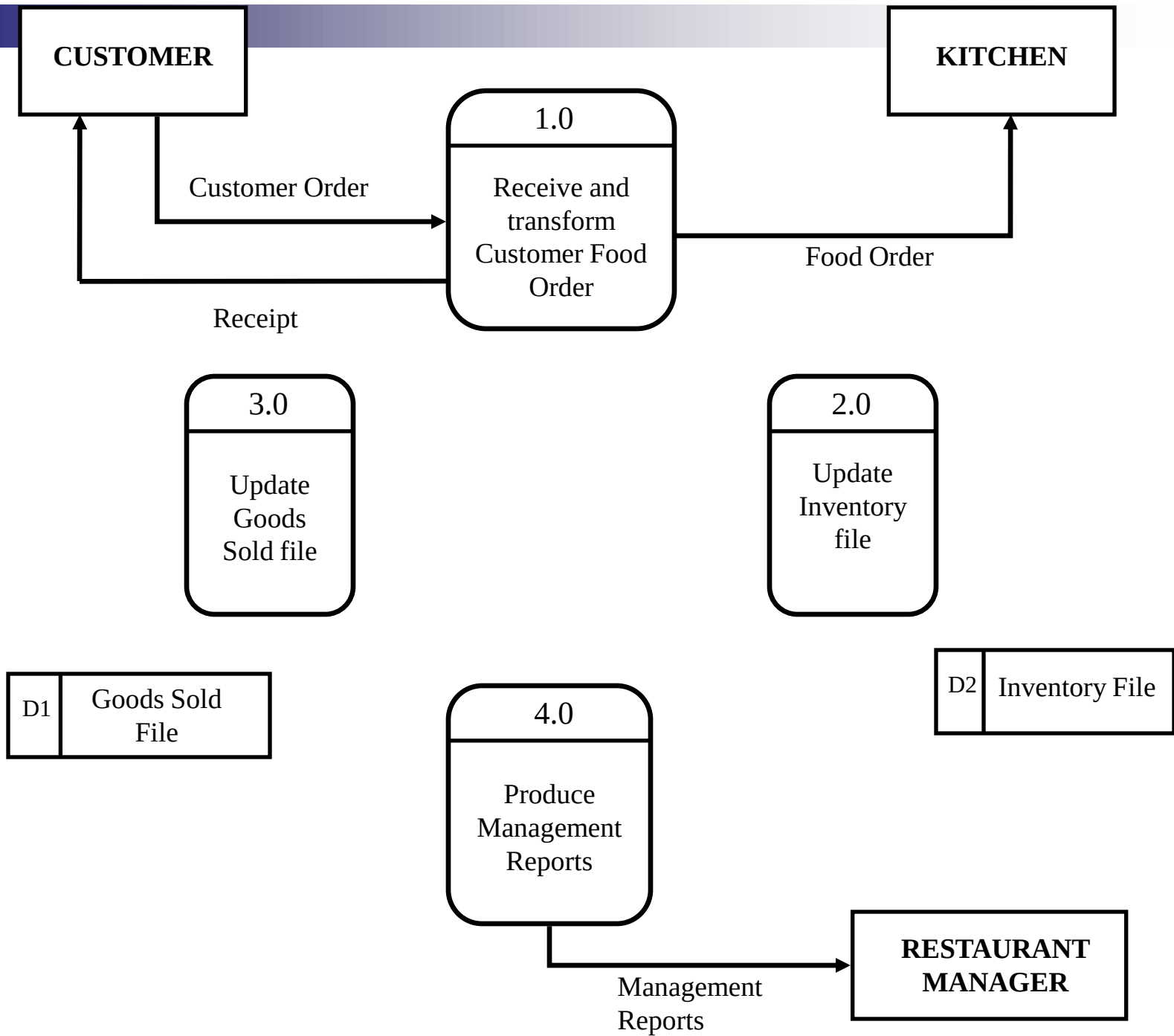
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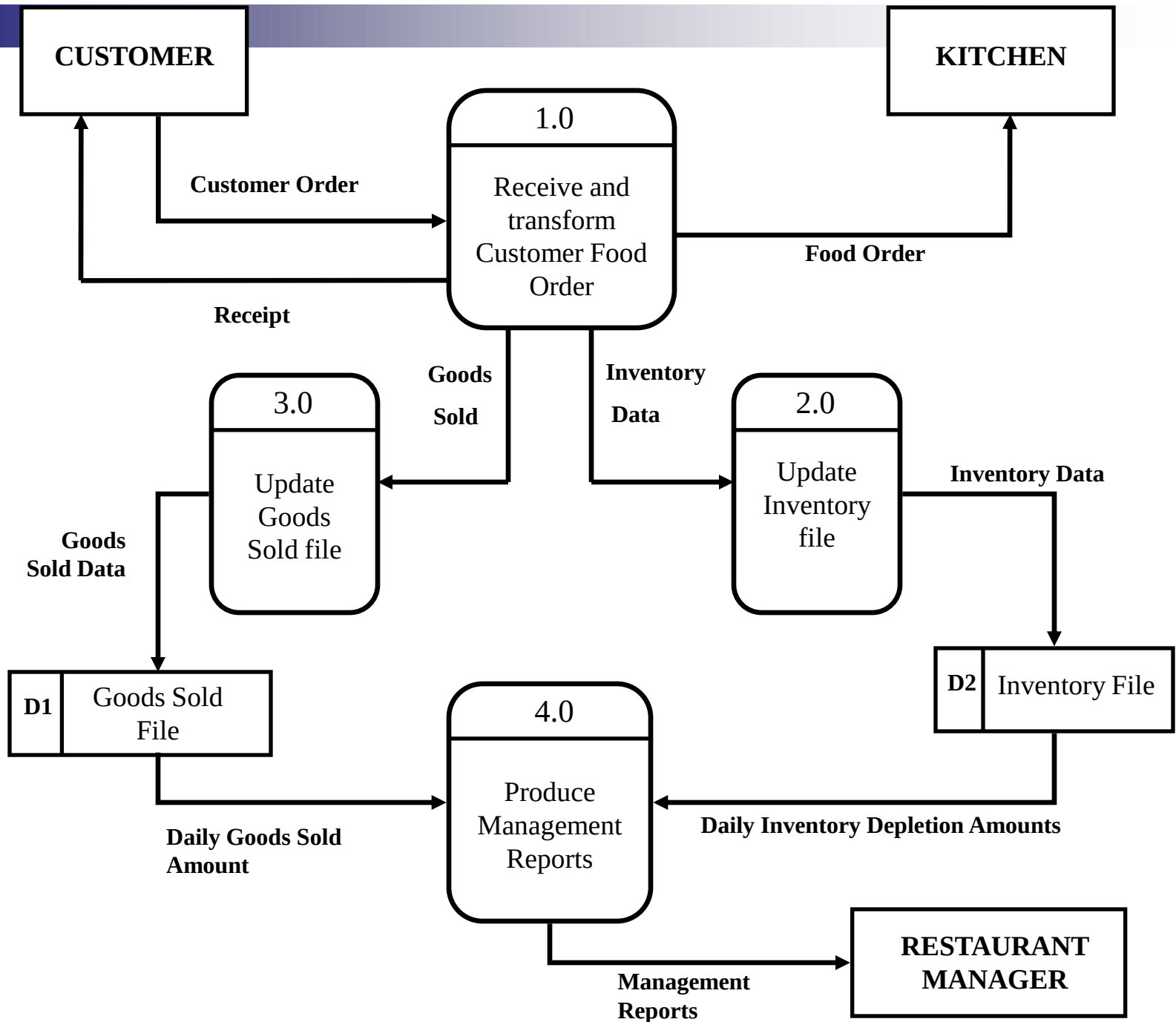
Data Flow Diagrams



Constructing a Level 1 diagram

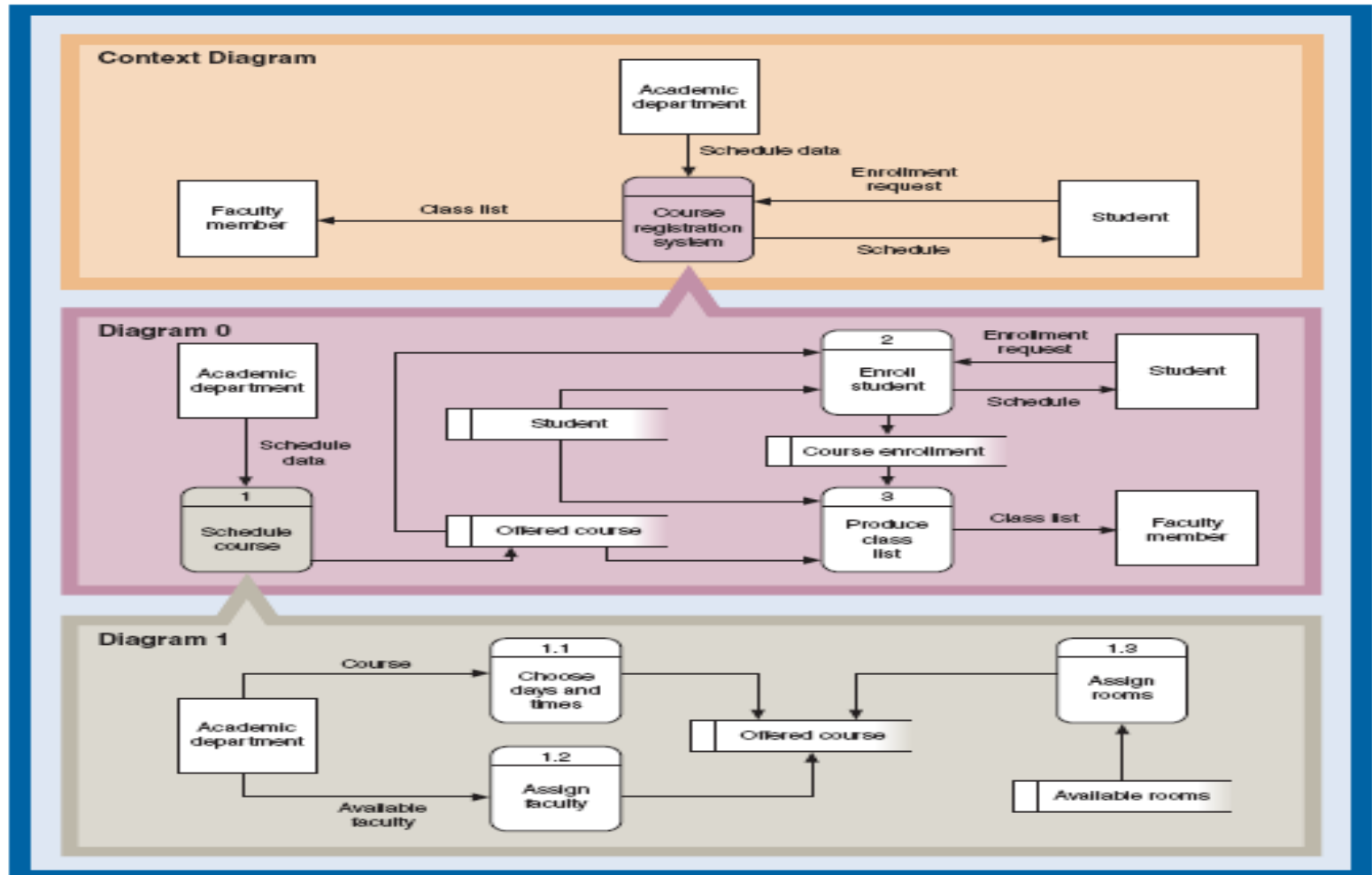
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Data Flow Diagrams

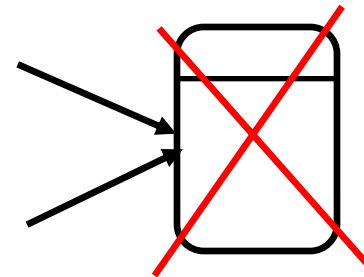
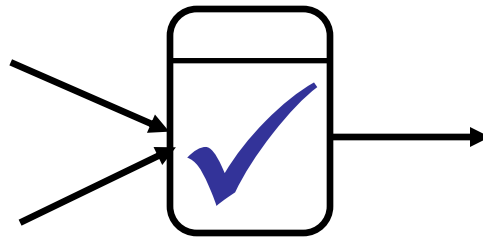
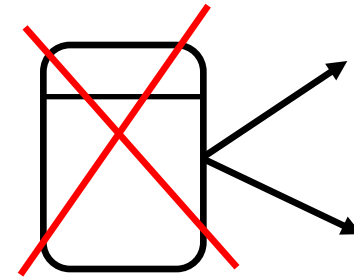
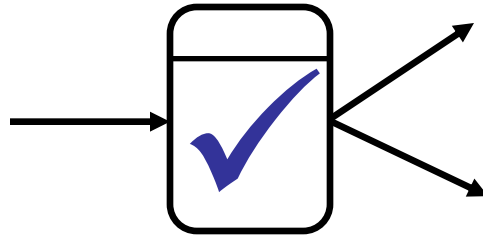


In this example, having constructed our Level 1 diagram it is decided there is no need to further breakdown the data flow processes within this organisation. Data flow diagram is therefore complete.

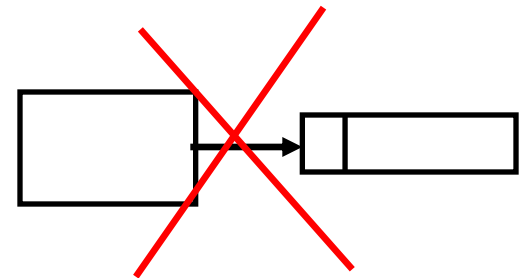
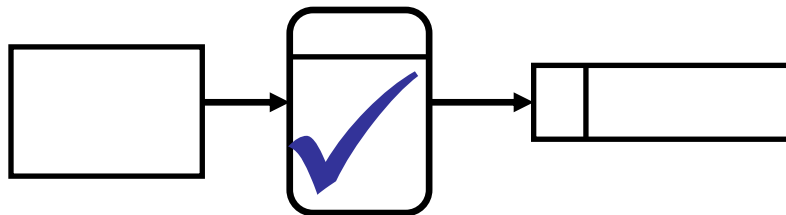
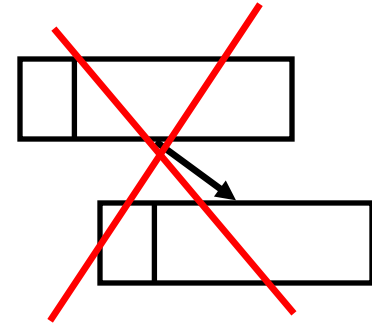
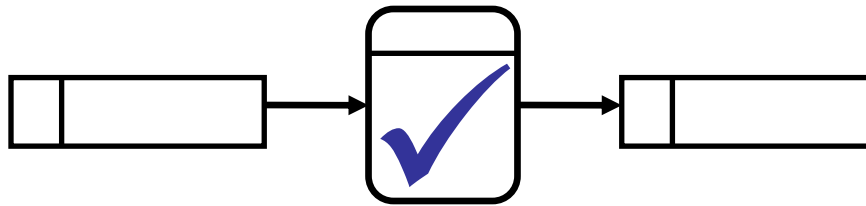
DFD for Course Registration System



Do's and Don'ts when constructing a Data Flow Diagram

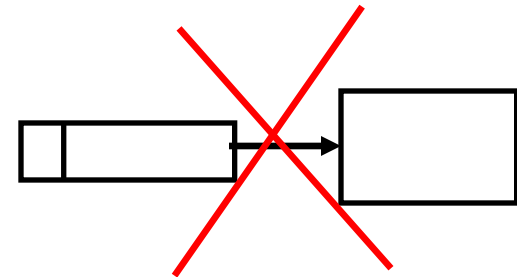
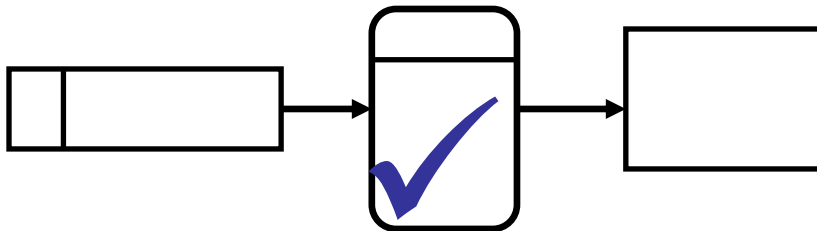
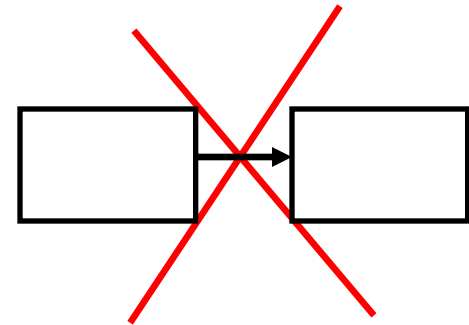
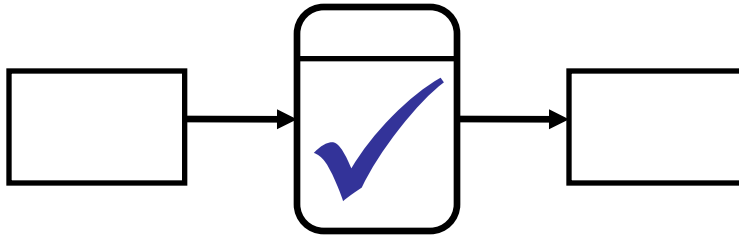


Do's and Don'ts when constructing a Data Flow Diagram



Do's and Don'ts when constructing a Data Flow Diagram

Do's and Don'ts when constructing a Data Flow Diagram





Balancing a DFD

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F The data that flow into the process and the data that flow out to the process need to be match when the process is split into in the next level. **This is known as balancing a DFD.**

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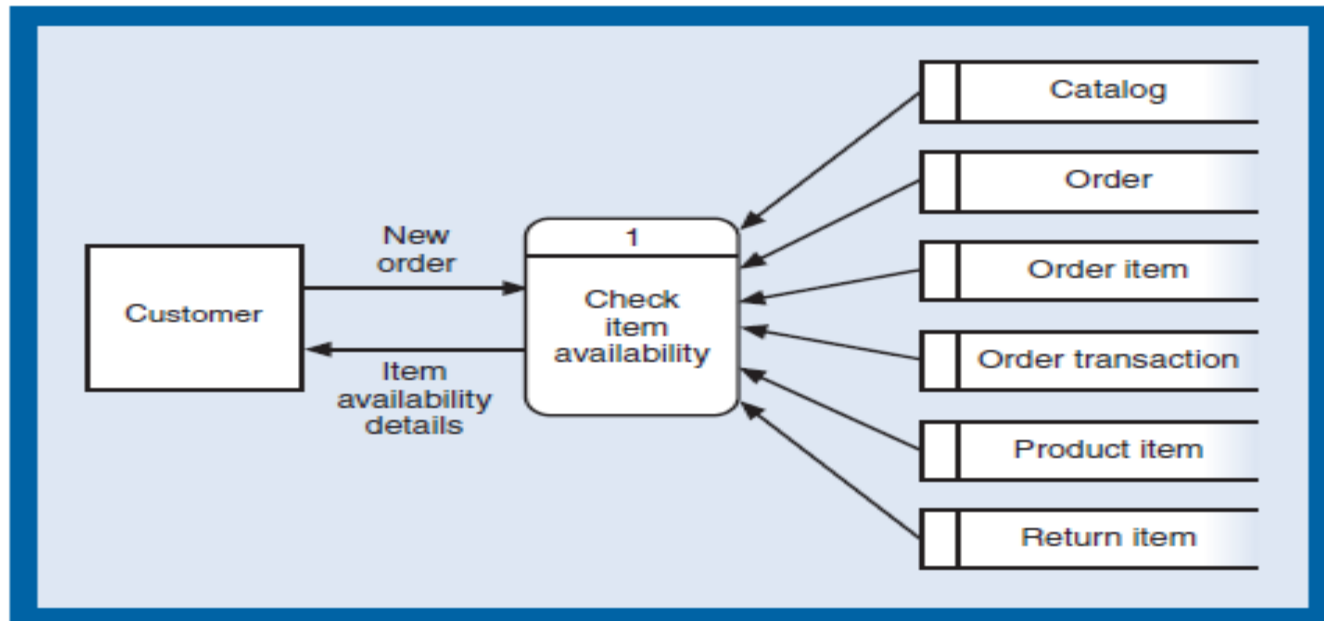
D Rules for Balancing

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- All data that flows into a process must flow out of the process or be used to generate data that flows out of the process.
- All data that flows out of a process must have flowed into the process or have been generated from data that flowed into the process.

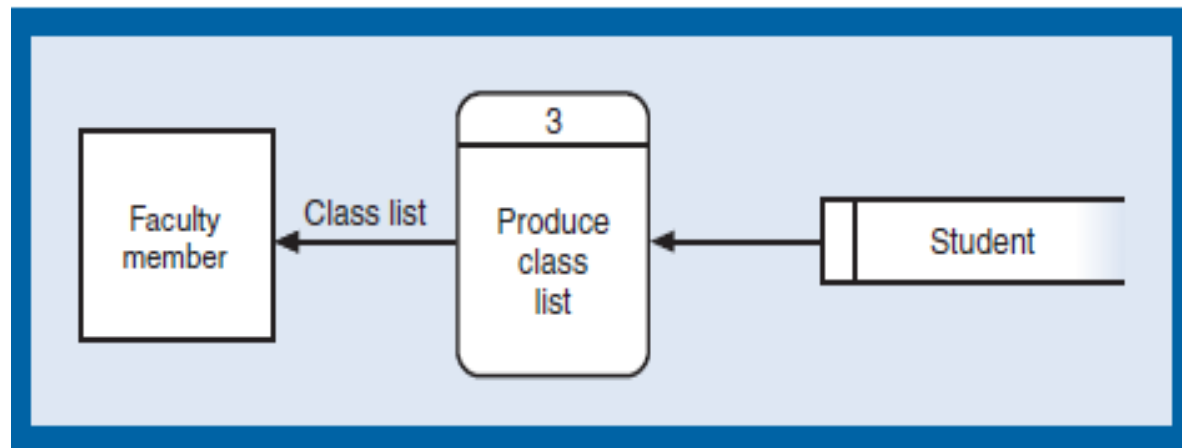
Black hole

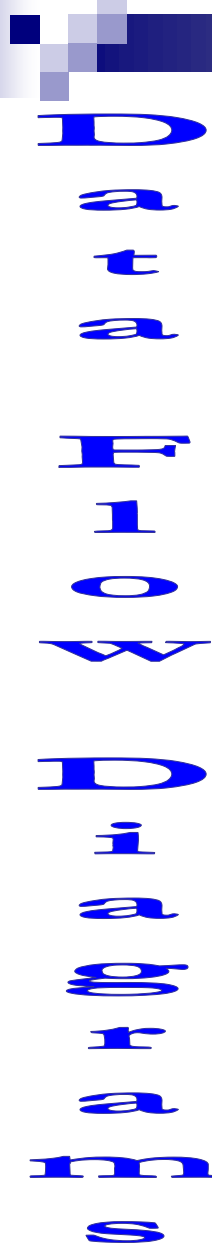
- A process or data store with a data input that is never used to produce a data output.



Magic/Miracle

- A process with an impossible data output—
a miracle
- A process or data store with a data element
that is created out of nothing





Documentation of DFD Components (Data Dictionary)

- Process Description
 - Structured English
 - Decision Tables
 - Decision Trees
- Data Flow Definitions
- Data Element Definition
- Data Store Definition

Exercise:

Precision Tools sells a line of high-quality woodworking tools. When customers place orders on the company's Web site, the system checks to see if the items are in stock, issues a status message to the customer, and generates a shipping order to the warehouse, which fills the order. When the order is shipped, the customer is billed. The system also produces various reports.

- Draw a context diagram for the order system
- Draw Level 1 DFD diagram for the order system

Identify Entities, Process, Data Stores & Data Flow

■ Entities

- ☐ Customer
- ☐ Warehouse
- ☐ Accounting

■ Processes

- ☐ 1.0 Check Status
- ☐ 2.0 Issue Status Messages
- ☐ 3.0 Generate Shipping Order
- ☐ 4.0 Manage Accounts Receivable
- ☐ 5.0 Produce Reports

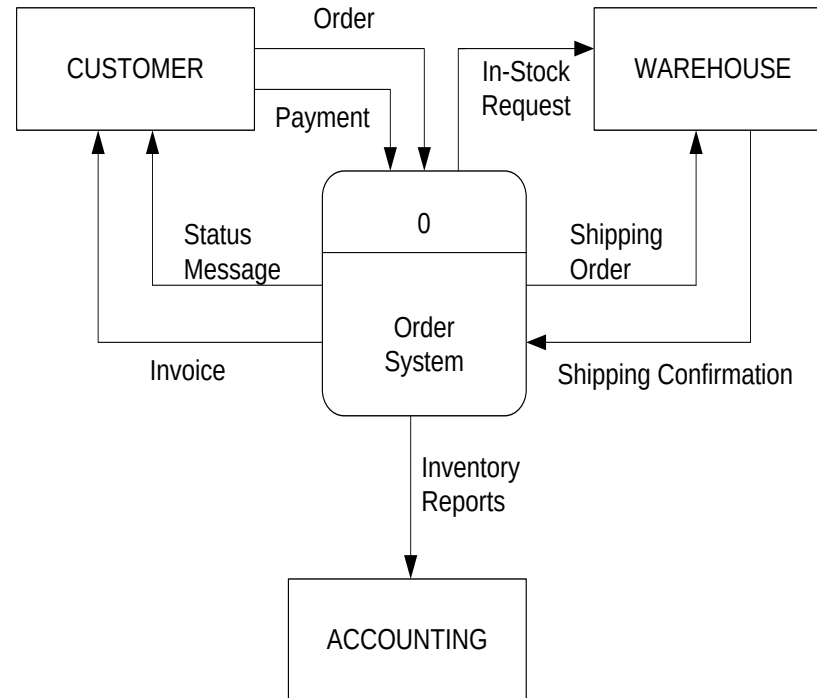
■ Data Stores

- ☐ D1 Pending Orders
- ☐ D2 Accounts Receivable

■ Data Flows

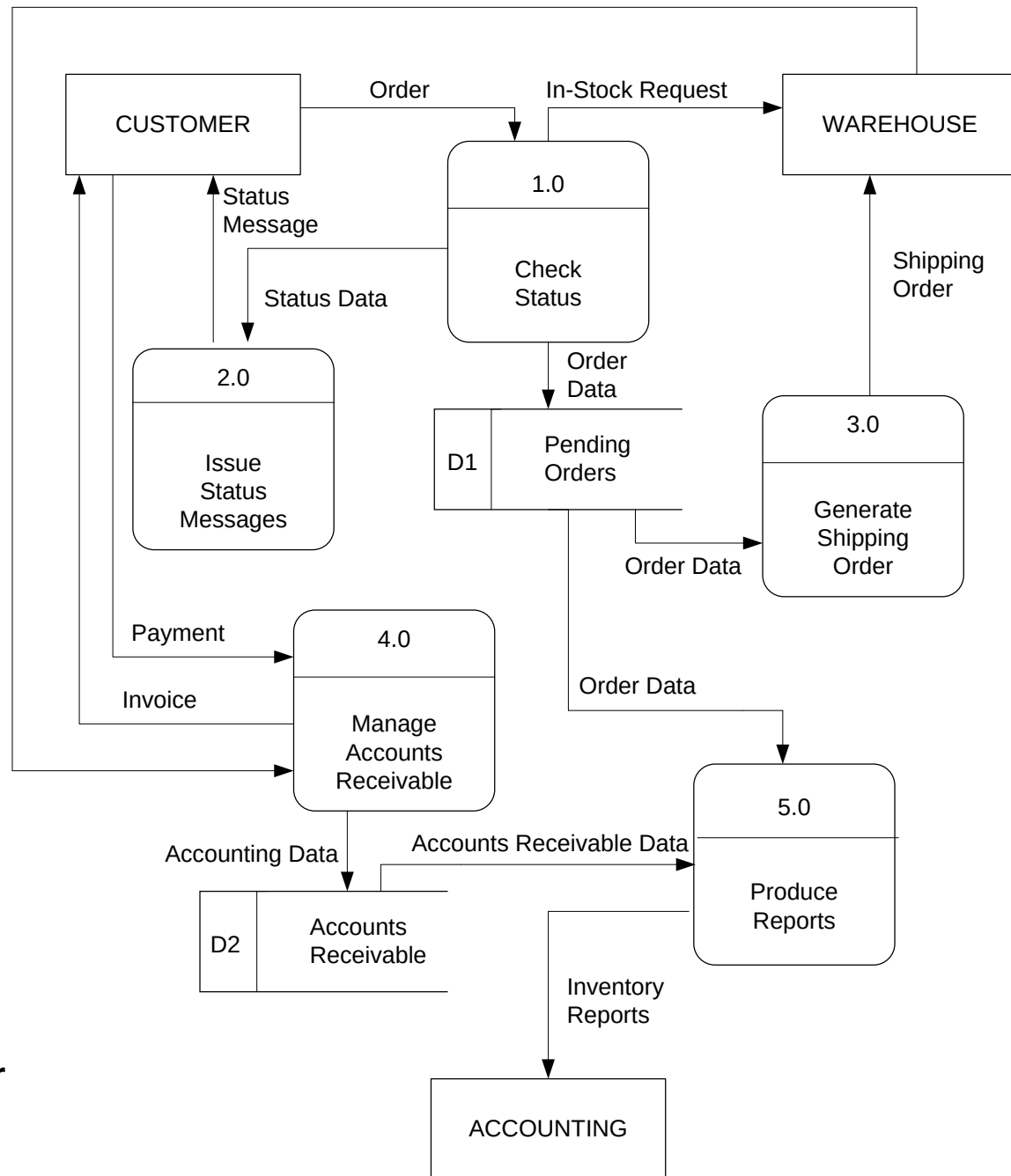
- ☐ Order
 - ☐ In-Stock Request
 - ☐ Order Data
 - ☐ Status Data
 - ☐ Status Message
 - ☐ Shipping Order
 - ☐ Order Data
 - ☐ Invoice
 - ☐ Shipping Confirmation
 - ☐ Payment
 - ☐ Accounting Data
 - ☐ Accounts Receivable Data
 - ☐ Order Data
 - ☐ Inventory Reports
- 1.0
- 2.0
- 3.0
- 4.0
- 5.0

Data Flow Diagrams



**Context Diagram of
Order System**

Data Flow Diagrams



Level-1 of Order System

Exercise:

□ Using the example of a retail clothing store in a mall, list relevant data flows, data stores, processes, and source/sinks. Draw a context diagram and level -1 diagram that represents selling system at the store.

□ Processes : POS, transform customer purchase, update goods/inventory/sales total file, produce management reports.

□ Entities : Customer, Store Manager

□ Data Stores : goods/inventory/sales total files



Steps in constructing a Functional Model

- ± Identify the inputs and outputs in the system.
- ± Draw Data Flow Diagrams.
- ± Describe Functionalities.
- ± Identify Constraints, if any.
- ± Identify Performance Requirements, if any.

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Identifying Data Flows

During the analysis stage of a project it is important to find out how data flows through a system:

- ± Where does the data originate
- ± What processing is performed on it and by whom
- ± Who uses the data
- ± What data is stored and where
- ± What output is produced and who receives it

Draw Data Flow Diagram

- ± Start with Top-level DFD.
- ± Break into smaller, detailed versions of DFD as described previously.

Describe Functionalities

- ± Can be describe in any form/language.
- ± For e.g. in plain English, pseudocode, flowcharts and so on.
- ± Emphasis should be on *What to do* rather than *How to do* it.

Describe Functionalities

Contd..

± **Requirement:** Book Return

± **Input:** Book ID

± **Output:** Status of operation.

± **Steps:**

± 1. Validate book id, if invalid perform error routine and stop execution.

± 2. Ensure that book is issued to member who is returning it.

± and so on...

Identify Constraints

± Constraints represent business rules and are very critical for an error-free and correct operation.

± For e.g in case of library system:

→ 1 member can issue at the most 1 book at any given time.

→ 1 book must belong to 0 or 1 member at any given time.

→ Books should be returned within 5 days from issue.

Identify performance requirements

± This requirement is useful in most modern applications using client/server architecture or web architecture.

± For e.g. Transaction should not take more than 10 milliseconds may mean either complete end to end transaction or just posting of entries in database.

Review Questions?

- Draw three levels of DFD for a supermarket App. [EQ](Refer class notes for problem statement)
- Draw DFD(upto 2 levels) for a payroll system. How do you map DFD to a structure chart?[EQ]
- Draw the DFD for order entry system (upto 2 levels) and describe the procedure for its validation. (DFD Rules are followed) [EQ]

Review Questions

- Explain DFD with one example. [EQ]
- Define (refer satzinger)
 - ☐ Context Diagram
 - ☐ DFD fragment
 - ☐ Event partitioned diagram
 - ☐ Magic & Black hole w.r.t DFD
- Draw two levels of DFD for stock/inventory management system[EQ].